

MT VERNON OUTLAND, IL, USA

WMO: 724335

Lat: 38.323N

Lon: 88.858W

Elev: 146

StdP: 99.58

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 93894

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.7	-12.0	-20.5	0.6	-13.5	-17.3	0.8	-10.4	11.0	11.4	9.6	10.7	1.7	280	0.451

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	34.8	24.9	33.1	24.6	32.3	24.2	26.8	32.1	25.8	31.5	25.1	30.7	3.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.7	30.3	24.0	19.2	29.5	23.0	18.1	28.2	84.5	32.3	80.8	31.6	76.9	30.6	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	8.0	7.0	DB	-20.3	36.7	4.2	1.7	-23.3	37.9	-25.7	38.8	-28.0	39.8	-31.0	41.0	(4)
(5)			WB	-20.4	27.6	4.2	1.3	-23.4	28.6	-25.9	29.3	-28.3	30.1	-31.3	31.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.0	-0.4	2.0	7.0	13.1	18.3	23.4	25.1	24.3	20.3	13.5	6.7	1.9
	DBStd	10.24	6.61	6.25	5.98	5.06	4.44	3.39	3.04	2.97	4.10	5.10	5.31	5.97
	HDD10.0	1123	327	233	131	26	1	0	0	0	0	22	126	257
	HDD18.3	2670	580	458	352	167	58	4	0	1	27	166	348	508
	CDD10.0	2217	6	9	39	119	259	401	467	444	308	130	28	7
	CDD18.3	723	0	0	2	11	58	156	208	188	85	16	1	0
	CDH23.3	7393	0	0	15	104	543	1636	2178	1860	876	174	7	0
	CDH26.7	3073	0	0	1	11	157	703	996	816	348	42	1	0
Wind	WSAvg	2.5	2.9	2.9	3.1	3.3	2.5	2.2	2.0	1.9	1.8	2.2	2.7	2.7
Precipitation	PrecAvg	1047	59	63	101	108	116	94	94	80	80	73	103	79
	PrecMax	1433	175	163	239	238	327	247	236	228	213	254	333	258
	PrecMin	763	1	7	25	44	19	5	22	8	12	7	13	14
	PrecStd	173	41	37	52	49	67	51	56	48	49	50	62	49
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.8	20.8	25.9	27.8	32.3	35.9	37.6	37.2	34.8	30.1	23.9	19.0
		MCWB	14.9	15.7	17.8	19.1	23.0	24.0	25.5	24.9	22.9	22.0	16.4	15.7
	2%	DB	15.9	17.5	22.5	26.3	30.2	33.8	34.9	34.1	32.6	27.6	20.4	16.3
		MCWB	13.3	13.0	15.6	18.1	22.5	24.2	25.8	24.5	23.3	19.2	15.7	13.2
	5%	DB	12.8	15.0	19.8	24.1	28.2	32.5	33.0	32.6	31.0	25.1	17.9	13.7
		MCWB	10.3	11.3	13.9	17.1	21.2	23.9	25.4	24.5	22.5	18.0	13.9	10.8
	10%	DB	9.1	12.3	17.3	22.4	27.0	31.1	32.1	31.2	28.0	22.6	16.1	11.1
		MCWB	6.9	8.7	12.5	16.1	20.2	23.3	24.8	23.7	20.6	17.0	12.0	8.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.4	16.9	18.7	21.1	24.6	26.7	28.5	27.7	25.4	23.3	18.6	16.7
		MCDB	17.5	19.6	24.2	26.2	29.9	32.1	33.4	32.3	31.9	28.6	22.9	18.5
	2%	WB	13.6	14.5	16.7	19.6	23.5	25.7	27.3	26.2	24.4	21.2	16.5	13.9
		MCDB	15.6	16.8	21.6	24.5	28.7	31.6	32.2	32.0	30.7	25.1	19.3	16.3
	5%	WB	10.7	11.8	14.9	18.3	22.2	24.9	26.3	25.3	23.2	19.5	14.8	11.2
		MCDB	12.4	14.5	18.7	22.9	27.2	30.6	31.8	31.1	28.7	23.3	18.0	13.5
	10%	WB	7.0	8.5	12.8	16.9	20.9	23.9	25.3	24.4	22.0	17.8	12.4	8.2
		MCDB	9.2	11.6	16.8	21.1	25.3	29.1	30.9	29.6	26.6	21.8	15.6	10.0
Mean Daily Temperature Range	5% DB	MDBR	9.2	9.7	11.2	12.3	11.9	12.2	11.6	12.2	13.5	13.0	11.1	9.0
		MCDBR	11.8	13.6	14.1	14.3	13.8	14.1	13.0	13.9	15.8	15.9	13.7	11.8
		MCWBR	10.1	10.6	9.4	8.7	7.3	6.4	5.8	6.2	7.4	8.8	9.8	9.9
	5% WB	MCDBR	10.7	12.0	12.5	11.9	11.9	11.7	11.5	11.9	13.4	12.7	11.8	10.6
		MCWBR	10.0	10.4	9.0	8.2	7.0	6.1	5.8	6.1	6.8	7.7	9.6	9.7
Clear-Sky Solar Irradiance	taub		0.311	0.324	0.363	0.399	0.434	0.437	0.465	0.464	0.408	0.358	0.320	0.306
	taud		2.458	2.405	2.361	2.260	2.222	2.232	2.163	2.153	2.302	2.438	2.488	2.480
	Ebn at Noon		870	902	896	879	852	846	818	809	837	852	852	848
	Edh at Noon		82	98	113	133	141	140	148	146	119	94	79	75
All-Sky Solar Radiation	RadAvg		2.01	2.75	3.74	4.94	5.55	6.41	6.21	5.64	4.71	3.43	2.32	1.70
	RadStd		0.20	0.30	0.35	0.29	0.48	0.41	0.40	0.31	0.32	0.36	0.26	0.16

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (36 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+78

Nomenclature: See separate page